

APPLICATION SECTION

THE ANALYSIS OF AN UNUSUAL 2^4 FRACTIONAL FACTORIAL DESIGN

L.M. Haines	Department of Statistics and Biometry, University of
A.A. Rayner	Natal, P.O. Box 375, Pietermaritzburg 3200, South
and	Africa.
I.M. Gravett:	Ruakura Agricultural Research Centre, Private Bag,
	Hamilton, New Zealand.

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Summary:	This article discusses the analysis of an unusual 2^4 factorial design which is non-orthogonal due to the misapplication of certain treatment combinations. A linear model not of full rank is formulated to describe the design, some unusual features are noted, and the model is then used to identify appropriate reduced models of full rank for the analysis of the data. Further, certain linear relationships between the parameter estimates of the candidate reduced models are noted and these are explained in terms of the linear model not of full rank as being induced either by the model or by the data.
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1. INTRODUCTION

1.1 THE DATA

We consider here the analysis of an unusual 2^4 factorial design which is non-orthogonal due to the misapplication of certain treatments. The

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experiment concerned was conducted on potatoes by H. Malicorner at Pagny-La-Ville, Côte d'Or, France in 1955. The factors were:

Varieties (V)	: Voran (v) <i>vs.</i> Osbote
Planting date (P)	: late (p) <i>vs.</i> early
Calcium (C)	: calcium (c) <i>vs.</i> no calcium
Form of potash (K)	: sulphate (k) <i>vs.</i> chloride

and the yields of tubers in quintals per hectare (1 quintal = 100 kg) are shown in Table 1.

Table 1
Derandomized plan and yields of the 2⁴ experiment on potatoes

Replication 1				Replication 2			
Block 1(a)		Block 1(b)		Block 2(a)		Block 2(b)	
<i>pk</i> *(<i>p</i>) [†]	373	<i>vp</i>	402	<i>pk</i> *(<i>p</i>) [†]	384	<i>vp</i>	429
<i>v</i>	453	(1)	386	<i>v</i>	361	(1)	369
<i>vpc</i>	433	<i>pc</i>	333	<i>vpc</i>	399	<i>pc</i>	333
<i>c</i>	371	<i>vc</i>	392	<i>c</i>	343	<i>vc</i>	387
<i>vpk</i>	416	<i>pk</i>	356	<i>vpk</i>	392	<i>pk</i>	365
<i>k</i>	341	<i>vk</i>	451	<i>k</i>	348	<i>vk</i>	383
<i>pc</i> *(<i>pck</i>) [†]	374	<i>vpck</i>	437	<i>pc</i> *(<i>pck</i>) [†]	336	<i>vpck</i>	360
<i>vck</i>	429	<i>ck</i>	375	<i>vck</i>	363	<i>ck</i>	373

[†] The treatment *p* was accidentally applied as *pk* and the treatment *pck* was accidentally applied as *pc*.

The effect of the form of potash was deemed to be of particular importance. The design was clearly intended to be a standard 2⁴ arrangement with the four-factor interaction, VPCCK, completely confounded with blocks. However, because of the treatment misapplication, the design is seen to be a $\frac{7}{8}$ replicate of a 2⁴ factorial, with the $\frac{1}{8}$ fraction {*p, pck*} omitted, but with the $\frac{1}{8}$ fraction {*pc, pk*} occurring twice in each replication.

1.2 A PRELIMINARY LOOK AT THE STATISTICAL ANALYSIS

We consider initially a least squares analysis with the blocks ignored and with the model for any plot yield (y) expressed in terms of main effects and interactions as

$$(1.1) E(y) = \mu \pm V \pm P \pm C \pm K \pm VP \pm VC \pm PC \pm VK \pm PK \pm CK \pm VPC \pm VPK \pm VCK \pm PCK \pm VPCK$$

(cf. Rayner, 1969, pp. 509-511; John, 1971, pp. 124-126). Specifically, μ is a common parameter, while the coefficients of the main effects V, P, C and K are $+1$ for the upper level of the factor and -1 for the lower level, and the coefficients of any interaction are given by the product of the coefficients of the main effects of the factors concerned. Without the treatment misapplication, the model matrix U for (1.1) would be of order 32×16 and rank 16, and the effects in (1.1) would represent 15 degrees of freedom for treatments (eliminating μ). With the misapplication, however, there are only 13 degrees of freedom available, and the rank of U drops to 14. Thus there must be two linearly independent exact linear relations across the columns of U .

In these circumstances we might utilize a linear model program, *e.g.* Genstat 5 (Payne *et al.*, 1987), which detects any such exact linear relationships and automatically breaks them by deleting the last columns of U involved in each. This is equivalent to setting the parameters corresponding to the deleted columns equal to zero (dropping them from the model). In this way a model not of full rank is reduced to one of full rank. However, such an automatic procedure may not be suitable. For example, with the order of effects as in (1.1), it is found that the interactions VPK and $VPCK$ are automatically dropped from the model, whereas it may be preferred to drop VPC (which does not involve K) rather than VPK .

Alternatively, $VPCK$ and a three-factor interaction may be deliberately chosen to be dropped from the model by deleting the corresponding columns of U . However, the choice of the three-factor interaction is not free. Thus, with the choice of VPC and $VPCK$, one finds that VPK is also dropped from the model. On the other hand, the choice of $VPCK$ and VPC is not

accompanied by the involuntary dropping of any other effect. The relevant theoretical considerations are discussed in section 2.

1.3 POST-ANALYTICAL OBSERVATIONS

It was noticed that with the full model (see below) the least squares estimates of the treatment effects when different pairs of interactions are dropped from the model are in certain cases linearly related. This is discussed in section 3.

2. PROPERTIES OF THE DESIGN AND THE ANALYSIS OF THE DATA

2.1 THE FULL MODEL

The linear model for the design is considered in the form:

$$(2.1) \quad \underline{y} = W\underline{\gamma} + X\underline{\beta} + \underline{\varepsilon} ,$$

where $\underline{y}$ is the 32×1 vector of observations, ordered according to the entries for blocks 1a, 1b, 2a, and 2b of Table 1 and $\underline{\varepsilon}$ is a 32×1 random vector assumed $N(0, \sigma^2 I)$.

For convenience in the ensuing discussions the following features which depart from the usual have been built into the model:-

(i) The term $W\underline{\gamma}$ is the blocks component of the model, with $\underline{\gamma}' = [\gamma_1 \gamma_2 \gamma_3]'$ corresponding to three single degrees of freedom for blocks: γ_1 represents the replication contrast, γ_3 the contrast between block pairs (a vs. b), and γ_2 the difference between the block-pair contrast over the two replications. It is clear from Table 1 that with this parameterization γ_3 is the only block degree of freedom confounded with treatments. The rows of W corresponding to blocks 1a, 1b, 2a, and 2b are, respectively, $[1 \ 1 \ 1]$, $[1 \ -1 \ -1]$, $[-1 \ -1 \ 1]$, and $[-1 \ 1 \ -1]$.

(ii) The common parameter, μ , which usually precedes the block effects, here follows them and is amalgamated with treatments in $Z\underline{\beta}$. The remaining elements of $\underline{\beta}$ represent the standard reparameterization to main effects and interactions as in (1.1), but in the particular order:

$$(2.2) \quad \underline{\beta}' = [\mu \ V \ P \ VP \ CK \ VCK \ PCK \ VPCK \\ C \ VC \ PC \ VPC \ K \ VK \ PK \ VPK].$$

It follows that Z is the same as U except for rearranged columns, and therefore that model (2.1) is not of full rank. For convenience we shall refer to $\underline{\beta}$ as the treatment parameters.

2.2 ALIASING AND ESTIMABILITY

2.2.1 Block parameters

A treatment combination in block 1a is repeated in block 2a so that the corresponding rows of Z are identical and thus it follows that $[1 \ 1 \ 0 \ \underline{0}'] \in \mathcal{R}[W \ Z]$, where $\mathcal{R}$ denotes row-space. Similarly, since a treatment combination in block 1b is repeated in block 2b, it follows that $[-1 \ 1 \ 0 \ \underline{0}'] \in \mathcal{R}[W \ Z]$, and since the repeated fraction $\{pc, pk\}$ occurs in blocks 1a and 1b that $[0 \ 1 \ 1 \ \underline{0}'] \in \mathcal{R}[W \ Z]$. Thus, overall,

$$\begin{bmatrix} 1 & 0 & 0 & \underline{0}' \\ 0 & 1 & 0 & \underline{0}' \\ 0 & 0 & 1 & \underline{0}' \end{bmatrix} \subset \mathcal{R}[WZ]$$

and hence the parameters γ_1, γ_2 , and γ_3 are estimable with respect to model (2.1) (Theorem 1.1 of Appendix 1).

2.2.2 Treatment parameters

Let Z_c represent the 16×16 model matrix for the minimal complete 2^4 design (Raktoe, Hedayat, and Federer, 1981, p.9) with columns corresponding to the order (2.2), and let

$$Z_0 = \begin{bmatrix} z'_p \\ z'_{pck} \end{bmatrix} = \begin{bmatrix} + & - & + & - & + & - & + & - & - & + & - & + & - & + & - \\ + & - & + & - & + & - & + & - & + & - & + & - & + & - & + \end{bmatrix},$$

where z'_p and z'_{pck} represent the rows from Z_c corresponding to the treatment combinations p and pck of the omitted fraction. Since the rows of Z_c are orthogonal,

$$(2.3) \quad Z Z'_0 = 0.$$

This equation describes the aliasing involving the parameters, $\underline{\beta}$ (see Definition 1.1 of Appendix 1).

Further, let $\underline{\beta}$ be partitioned as $\begin{bmatrix} \underline{\beta}_1 \\ \underline{\beta}_2 \end{bmatrix}$, where

$\underline{\beta}'_1 = [\mu \ V \ P \ VP \ CK \ VCK \ PCK \ VPCK]$ and

$\underline{\beta}'_2 = [C \ VC \ PC \ VPC \ K \ VK \ PK \ VPK]$, with Z conformably partitioned as

$Z = [Z_1 \ Z_2]$. Then it follows on post-multiplying equation (2.3) by the matrix $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$ that the aliasing can be equivalently defined by

$$(2.4) \quad Z_1 \underline{a} = \underline{0} \quad \text{and} \quad Z_2 \underline{a} = \underline{0}$$

where $\underline{a}' = [+ - + - + - + -]$. Thus there are two disjoint sets of elements of $\underline{\beta}$ involved in aliasing, viz. $T_1 = \{\mu, V, P, VP, CK, VCK, PCK, VPCK\}$ and $T_2 = \{C, VC, PC, VPC, K, VK, PK, VPK\}$, and these are the alias sets of the design (John, 1971, pp. 163 and 171). It is seen that the order (2.2) is the order of these alias sets (John, 1971, pp. 150 and 161).

It is interesting to observe that the present irregular fraction can be regarded as complementary to the omitted $\frac{1}{8}$ fraction, $\{p, pck\}$, through the relation (2.3). Thus for example the estimable functions for the design, $\{p, pck\}$, are non-estimable functions of the present design and *vice versa* (Theorem 1.1 of Appendix 1). Further the omitted $\frac{1}{8}$ fraction is regular, with alias sets given by its defining contrast

$$I = -V = P = -VP = CK = -VCK = PCK = -VPCK$$

and defining contrast subgroup

$$C = -VC = PC = -VPC = K = -VK = PK = -VPK.$$

Since all the individual parameters of $\underline{\beta}$ are involved in aliasing they are not themselves estimable (Theorem 1.2 of Appendix 1). However, since the elements of γ are estimable, $\mathcal{R}[W \ 0] \subset \mathcal{R}[WZ]$ and hence

$\mathcal{R}[WZ] = \mathcal{R} \begin{bmatrix} W & 0 \\ 0 & Z \end{bmatrix}$; thus a linear function of the parameters $\underline{\beta}$, say $\underline{\theta}' \underline{\beta} = [\underline{\theta}'_1 \ \underline{\theta}'_2] \begin{bmatrix} \underline{\beta}_1 \\ \underline{\beta}_2 \end{bmatrix}$, is estimable if and only if $\underline{\theta}' \in \mathcal{R}(Z)$ or

equivalently, from (2.4), if and only if the subvectors $\underline{\theta}_1$ and $\underline{\theta}_2$ are both orthogonal to $\underline{a}$. We note that $\text{rank } [WZ] = \text{rank } (W) + \text{rank } (Z) = 17$.

2.3 THE NORMAL EQUATIONS

Since model (2.1) is not of full rank the corresponding normal equations,

$$(2.5) \quad \begin{bmatrix} W'W & W'Z \\ Z'W & Z'Z \end{bmatrix} \begin{bmatrix} \hat{\gamma} \\ \underline{\beta}^0 \end{bmatrix} = \begin{bmatrix} W'y \\ Z'y \end{bmatrix},$$

are necessarily indeterminate, the matrix $S = \begin{bmatrix} W'W & W'Z \\ Z'W & Z'Z \end{bmatrix}$ of order 19 being singular with rank 17. Note that $\hat{\gamma}$ represents the unique estimate of the block parameters, $\underline{\gamma}$, whereas $\underline{\beta}^0$ represents a non-unique solution for the treatment parameters, $\underline{\beta}$.

To examine the form of S we introduce

$$Z_a = \begin{bmatrix} \underline{z}'_{pk} \\ \underline{z}'_{pc} \end{bmatrix} = \begin{bmatrix} + & - & + & - & - & + & - & + & - & + & - & + & + & - & + & - \\ + & - & + & - & - & + & - & + & + & - & + & - & - & + & - & + \end{bmatrix},$$

where $\underline{z}'_{pk}$ and $\underline{z}'_{pc}$ represent the rows of Z_c corresponding to the treatment combinations pk and pc , respectively, of the repeated fraction. Then $Z'Z = 2[Z'_c Z_c - Z'_0 Z_0 + Z'_a Z_a]$ (2 replications). But $Z'_c Z_c = 16I_{16}$, $Z'_0 Z_0 = 2 \begin{bmatrix} L & L \\ L & L \end{bmatrix} \otimes I_2$, and $Z'_a Z_a = 2 \begin{bmatrix} L & -L \\ -L & L \end{bmatrix} \otimes I_2$, where I_n repre-

sents the identity matrix of order n , $L = \begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix}$, and $\otimes$ de-

notes the left Kronecker product, so that $Z'Z = \begin{bmatrix} 32I_4 & -8L \\ -8L & 32I_4 \end{bmatrix} \otimes I_2 =$

$\begin{bmatrix} Z'_1 Z_1 & 0 \\ 0 & Z'_2 Z_2 \end{bmatrix}$. The quasi-diagonal structure of $Z'Z$ is determined by the particular combination of omitted and repeated $\frac{1}{8}$ fractions in the present 2^4 design and would not necessarily occur for other choices of these fractions. The normal equations in numerical form are shown in

Table 2. From the form of S it is seen that the treatment effects CK , VCK , PCK , as well as $VPCK$, are confounded with the block effect γ_3 .

2.4 REDUCED MODEL

The estimable functions of $\underline{\beta}$ discussed in section 2.2 are not, however, what we want to estimate in an analysis of the data. Rather it is important to estimate as many of the individual treatment parameters as possible. This can clearly be achieved if we reduce the model (2.1) to one of full rank by excluding one effect from set T_1 and one from T_2 . This is the mathematical solution; an alternative version is given in Appendix 2.

From the statistical point of view the effect deleted from T_1 should be $VPCK$ on grounds of marginality (Nelder, 1977; Aitkin, 1978), or simply because, being a four-factor interaction, it "may in many cases be confidently predicted to be small in comparison with the errors affecting [it]" (Yates, 1937). The interactions in T_2 of highest order are VPC and VPK and, since the factor K is of particular importance, it seems sensible to retain VPK and delete VPC . Thus the reduced model

$$(2.6) \quad \begin{cases} \underline{y} = W\underline{\gamma} + Z\underline{\beta} + \underline{\varepsilon} \\ VPC = 0, VPCK = 0 \end{cases}$$

is adopted.

The normal equations for this model are given by

$$(2.7) \quad S_{11} \begin{bmatrix} \hat{\underline{\gamma}} \\ \hat{\underline{\beta}}_{1,VPCK} \\ \hat{\underline{\beta}}_{2,VPC} \end{bmatrix} = X_1' \underline{y},$$

where X_1 is the matrix $[W \ Z]$ with the columns corresponding to $VPCK$ and VPC removed, $S_{11} = X_1' X_1$, $\hat{\underline{\beta}}_{1,VPCK}$ denotes the vector $\underline{\beta}_1$ with the element $VPCK$ deleted and $\hat{\underline{\beta}}_{2,VPC}$ the vector $\underline{\beta}_2$ with the element VPC deleted.

The aliasing structure (Raktoe, Pesotan, and Federer, 1980) for the present design under model (2.6) is given by

$$E \begin{bmatrix} \hat{\gamma} \\ \hat{\beta}_{1,VPCK} \\ \hat{\beta}_{2,VPC} \end{bmatrix} = \begin{bmatrix} \gamma \\ \beta_{1,VPCK} \\ \beta_{2,VPC} \end{bmatrix} + S_{11}^{-1} X_1' X_2 \begin{bmatrix} VPCK \\ VPC \end{bmatrix},$$

where X_2 represents the deleted columns. Thus it is readily seen that $E(\hat{\gamma}) = \gamma$ and the aliasing structure is

$$E \begin{bmatrix} \hat{\beta}_{1,VPCK} \\ \hat{\beta}_{2,VPC} \end{bmatrix} = \begin{bmatrix} \beta_{1,VPCK} + VPCK \underline{a}_{(-8)} \\ \beta_{2,VPC} + VPC \underline{a}_{(-4)} \end{bmatrix},$$

where $\underline{a}_{(-i)}$ denotes vector $\underline{a}$ with its i -th element deleted.

2.5 STATISTICAL ANALYSIS

The non-orthogonality of the design is revealed by inspection of S_{11}^{-1} (Table 3). Thus treatment set T_1 is non-orthogonal to blocks but orthogonal to T_2 , treatment set T_2 is orthogonal to both blocks and T_1 , and the parameters within each treatment set are non-orthogonal to one another. Hence the order of fitting the individual terms in the reduced model affects the sums of squares of the analysis of variance to the extent that, provided blocks are fitted first, only the order within the treatment sets T_1 and T_2 makes any difference. Within each set the marginality principle (Nelder, 1977; Aitkin, 1978) must clearly be considered of overriding importance, i.e. terms are fitted in the order: common parameter, main effects, two-factor interactions and three-factor interactions. Further, within each of T_1 and T_2 it is sensible, in view of the stated importance of the factor K , to fit terms involving K before the others (except μ). There are then only four sequences of effects within T_1 , and, independently, four sequences within T_2 , which need to be considered.

An analysis of variance based on fitting (2.6) with one of these admissible orderings of effects is presented in Table 4. Only the main effect of V is significant. (Computationally it makes no difference whether μ is fitted before or after the block parameters. In Table 4, as is commonly done, the sum of squares due to fitting μ has been omitted and also subtracted

from the total sum of squares.)

Table 3. The matrix $128(S_{11}^{-1})$ for model (2.6).

4	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
.	4	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	16	12	-12	12	-12	16	-16	16	.	.	.	.	.	.	.	.
.	.	12	16	-12	12	-12	16	-16	16	.	.	.	.	.	.	.	.
.	.	-12	-12	16	-12	12	-16	16	-16	.	.	.	.	.	.	.	.
.	.	12	12	-12	16	-12	16	-16	16	.	.	.	.	.	.	.	.
.	.	-12	-12	12	-12	16	-16	16	-16	.	.	.	.	.	.	.	.
.	.	16	16	-16	16	-16	24	-20	20	.	.	.	.	.	.	.	.
.	.	-16	-16	16	-16	16	-20	24	-20	.	.	.	.	.	.	.	.
.	.	16	16	-16	16	-16	20	-20	24	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	8	-4	4	4	-4	4	-4	.
.	.	.	.	.	.	.	.	.	.	-4	8	-4	-4	4	-4	4	.
.	.	.	.	.	.	.	.	.	.	4	-4	8	4	-4	4	-4	.
.	.	.	.	.	.	.	.	.	.	4	-4	4	7	-3	3	-3	.
.	.	.	.	.	.	.	.	.	.	-4	4	-4	-3	7	-3	3	.
.	.	.	.	.	.	.	.	.	.	4	-4	4	3	-3	7	-3	.
.	.	.	.	.	.	.	.	.	.	-4	4	-4	-3	3	-3	7	.

NB. The matrix $128(S_{11}^{-1})$ for model (3.1) differs only in the 7×7 trailing submatrix. If that for model (2.6) is partitioned as $\begin{bmatrix} A_{3 \times 3} & B_{3 \times 4} \\ C_{4 \times 3} & D_{4 \times 4} \end{bmatrix}$,

then that for model (3.1) is $\begin{bmatrix} D_{4 \times 4} & C_{4 \times 3} \\ B_{3 \times 4} & A_{3 \times 3} \end{bmatrix}$

Table 4. A typical analysis of variance table for the reduced model (2.6).

Source	DF	SS	MS	F
Blocks (ignoring treatments)	3	5468.593		
Treatment set T_1 (eliminating blocks) in order of fitting				
V	1	16516.532		27.96**
P	1	0.281		0.00
CK	1	437.760		0.74
VP	1	152.274		0.26
VCK	1	1295.320		2.19
PCK	1	500.521		0.85
	(6)	(18902.688)		
Treatment set T_2 in order of fitting				
K	1	63.281		0.11
C	1	850.669		1.44
VK	1	91.934		0.16
PK	1	0.578		0.00
VC	1	16.001		0.03
PC	1	40.716		0.07
VPK	1	1716.071		2.90
	(7)	(2779.250)		
Error	15	8861.939	590.796	
Total	31	36012.469		

The solution to the normal equations (2.7), i.e. the parameter estimates, $\hat{\gamma}$, $\hat{\beta}_{1,VPCK}$, and $\hat{\beta}_{2,VPC}$ for (2.6), is given in Table 5. (The right-hand column of this table, $\hat{\beta}_{2,VPK}$, will be considered in section 3).

Table 5. Parameter estimates for the alternative reduced models (2.6) and (3.1).

$\hat{\gamma}' = [12.4063 \quad 4.0938 \quad 10.0]$				
SET T_1		SET T_2		
	$\hat{\beta}_{1,VPCK}$		$\hat{\beta}_{2,VPC}$	$\hat{\beta}_{2,VPK}$
$\hat{\mu}$	393.1875	$\hat{C}$	0.5	-9.1875
$\hat{V}$	12.25	$\widehat{VC}$	-5.9375	3.75
$\hat{P}$	10.375	$\widehat{PC}$	4.1875	-5.5
$\widehat{VP}$	-7.3125	$\widehat{VPC}$	0.0 [†]	9.6875
$\widehat{CK}$	15.5	$\hat{K}$	4.0625	-5.625
$\widehat{VCK}$	-16.6875	$\widehat{VK}$	-5.625	4.0625
$\widehat{PCK}$	9.6875	$\widehat{PK}$	4.0	-5.6875
$\widehat{VPCK}$	0.0 [†]	$\widehat{VPK}$	-9.6875	0.0 [†]

[†] These zero entries are not parameter estimates, but $VPCK = VPC = 0$ and $VPCK = VPK = 0$ are parameters made zero in the respective models (see section 3.1).

The fact that V alone is significant and yet that $\hat{V}$ is of modest magnitude in Table 5 suggests that the principle of parsimony in model selection be applied. Two possibilities seem reasonable:

- i) The main-effects-only model: Here only C and K are non-orthogonal. The estimated effects are: $\hat{V} = 22.72 \pm 4.13$, $\hat{P} = -0.09 \pm 4.13$, $\hat{C} = -5.32 \pm 4.27$, $\hat{K} = 0.08 \pm 4.27$, and the MS for lack of fit is 472.35 with 9 DF, compared with the error MS 590.80 for the full-rank model.
- ii) The model with V , K (the factor of interest), and their interaction: In this case the analysis reduces to that for an unconfounded 2^2 design with 2 replications per block. The estimated effects are: $\hat{V} = 22.72$, $\hat{K} = 1.41$, $\widehat{VK} = -2.97$ (all ± 4.14), and the MS for lack of fit is 482.01 with 10 DF. A presentation of the results of the experiment in terms of this model is given in Table 6.

Table 6. Variety x Form of Potash interaction table of treatment means

	Osbote	Voran	Mean	Difference between varieties in favour of Voran
Chloride	355.6	407.0	381.3	51.4**
Sulphate	364.4	403.9	384.1	39.5**
Mean	360.0	405.4	382.7	45.4**
Difference between forms of potash in favour of sulphate	8.8 ^{NS}	-3.1 ^{NS}	2.8 ^{NS}	-11.9 ^{NS}

SE's and least significant values (LSV's) for the above table:

	<u>SE</u>	<u>LSV (5%)</u>	<u>LSV(1%)</u>
Body of table	8.27	17.04	23.05
Marginal means	5.85	12.05	16.30
Marginal differences	11.70	24.10	32.60
Mean differences	8.27	17.04	23.05

One may observe that the varietal difference is highly significant for both forms of potash, the yields for Voran being greater than for Osbote, but that the difference of these differences is not significant in accord with the fact that the interaction is non-significant.

3. LINEAR RELATIONSHIPS BETWEEN PARAMETER ESTIMATES

In section 2.4 the alternative reduced model

$$(3.1) \quad \begin{cases} \underline{y} = W\underline{\gamma} + Z\underline{\beta} + \underline{\varepsilon} \\ VPK = 0, VPCK = 0 \end{cases}$$

was considered as a possibility. The treatment parameter estimates for (3.1), viz. $\hat{\underline{\beta}}_{1,VPCK}$ for $\underline{\beta}_1$ with VPCK omitted and $\hat{\underline{\beta}}_{2,VPK}$ for $\underline{\beta}_2$ with VPK omitted, are shown in Table 5, $\hat{\underline{\beta}}_{1,VPCK}$ being the same for both (2.6) and (3.1). The estimates $\hat{\underline{\beta}}_{2,VPK}$ and $\hat{\underline{\beta}}_{2,VPC}$ are different, of course, but certain linear relationships between them are obvious to the eye; for example,

$$(3.2) \quad \hat{C}_{VPK} = \hat{C}_{VPC} + \widehat{VPK}_{VPC}$$

and

$$\hat{K}_{VPC} = \widehat{VK}_{VPK},$$

and these "coincidences" are explored here.

3.1 MODEL-INDUCED RELATIONSHIPS BETWEEN THE TREATMENT PARAMETER ESTIMATES FOR THE REDUCED MODELS (2.6) AND (3.1)

We begin by noting the connection between the treatment parameter estimates for the two reduced models and the solutions to the normal equations (2.5) for model (2.1) obtained by imposing corresponding constraints on the solutions (Searle, 1971, p.205). Specifically, the estimates of the treatment parameters for the reduced model with effects $E_1 \in T_1$ and $E_2 \in T_2$ excluded, i.e. $\hat{\underline{\beta}} = \begin{bmatrix} \hat{\underline{\beta}}_{1,E_1} \\ \hat{\underline{\beta}}_{2,E_2} \end{bmatrix}$, and the solution to the normal equations (2.5), say $\underline{\beta}^0 = \begin{bmatrix} \underline{\beta}_{1,E_1}^0 \\ \underline{\beta}_{2,E_2}^0 \end{bmatrix}$, obtained by imposing the non-estimable constraints $E_1^0 = 0$ and $E_2^0 = 0$, are numerically the same, except for the fact that $\underline{\beta}_{i,E_i}^0$ contains one more element than $\hat{\underline{\beta}}_{i,E_i}$, viz. the zero E_i^0 . Consequently, with the convention that $\hat{E}_i = E_i = 0$, Table 5 applies equally to constrained solutions and parameter estimates for reduced models. It follows that the concept of estimability, in the sense

that, for any estimable function $\underline{\theta}'\underline{\beta}$, $\underline{\theta}'\underline{\beta}^0$ is constant for any solution to (2.5) corresponding to constraints $E_1^0 = E_2^0 = 0$, extends likewise to $\underline{\theta}'\underline{\hat{\beta}}$ for any reduced model with $E_1 = E_2 = 0$.

Model-induced relationships between solutions to the normal equations (2.5) follow immediately from Theorem 1.3 of Appendix 1. In particular for set T_2 , since by (2.4) $\underline{a}'$ is an orthogonal complement of Z_2 ,

$$\underline{\beta}_{2,VPK}^0 = \underline{\beta}_{2,VPC}^0 + VPK_{VPC}^0 \underline{a},$$

which explains (3.2) for example. Another way of seeing this is to check the estimability of $C + VPK$ by the method of section 2.2.2; then from Table 5 $C^0 + VPK^0 = -9.1875$, constant for any solution to (2.5).

3.2 DATA-INDUCED RELATIONSHIPS BETWEEN THE TREATMENT PARAMETER ESTIMATES FOR (2.6) AND (3.1)

Relationships between the treatment parameter estimates for (2.6) and (3.1) which are not induced by the non-full-rank model were also noted, for example,

$$(3.3) \quad \hat{K}_{VPC} - \widehat{VK}_{VPC} = 0$$

and

$$(3.4) \quad \widehat{PCK}_{VPC} + \widehat{VPK}_{VPC} = 0.$$

From (2.8)

$$E \begin{bmatrix} \hat{\beta}_{1,VPCK} \\ \hat{\beta}_{2,VPC} \end{bmatrix} = \begin{bmatrix} I_7 & \underline{a}_{(-8)} \\ I_7 & \underline{a}_{(-4)} \end{bmatrix} \begin{bmatrix} \underline{\beta}_1 \\ \underline{\beta}_2 \end{bmatrix}$$

and, since $\begin{bmatrix} I_7 & \underline{a}_{(-8)} \\ I_7 & \underline{a}_{(-4)} \end{bmatrix} \underline{a} = \underline{0}$, $E(\hat{\beta}_{1,VPCK})$ and $E(\hat{\beta}_{2,VPC})$ are estimable. Considering (3.3) in particular

$$E(\hat{K}_{VPC} - \widehat{VK}_{VPC}) = K + VPC - VK + VPK,$$

which is estimable, has a zero estimate for the given $\underline{y}$. Conversely, if the estimable function

$$(3.5) \quad K + VPC - VK + VPK$$

has a zero estimate, then $\hat{K}_{VPC} - \widehat{VK}_{VPK}$ must also be zero.

Similarly it can be shown that (3.4) holds if and only if the estimable function

$$(3.6) \quad PCK + VPCK - VPC + VPK$$

has a zero estimate.

3.3 ESTIMABLE FUNCTIONS WITH ZERO ESTIMATES

The above observations lead naturally to a consideration of the set of all estimable functions which have zero estimates, or, equivalently, of the set of vectors

$$L = \{\underline{\theta} : \underline{\theta}'\underline{\beta} \text{ estimable and } \underline{\theta}'\underline{\beta}^0 = 0\},$$

where L is a subspace of dimension 13 of the vector space, R^{16} . It is interesting to investigate whether there exists a basis for the subspace, L , which involves vectors with small integer elements such as those of (3.5) and (3.6). To this end a systematic computer-based search for linearly independent vectors belonging to L which have four non-zero integer elements between -9 and +9 was implemented. This provided twelve such vectors and the corresponding estimable functions are the first twelve expressions in Table 7.

To complete the basis for L a vector with a non-zero element as the coefficient for μ is clearly required. One such vector was obtained by inspection and the corresponding estimable function is shown as the last entry of Table 7.

A more fundamental approach to the examination of estimable functions with zero estimates is to identify the underlying relationships between the observations, $\underline{y}$ or, more concisely, the vector of treatment totals, $\underline{t}$, where $\underline{t} = T\underline{y}$ with $T = [I_{16} \ I_{16}]$, i.e.

$$\underline{t}' = [t_{pk} \ t_v \ t_{upc} \ t_c \ t_{vpk} \ t_k \ t_{pc} \ t_{uck} \ t_{vp} \ t_{(1)} \ t_{pc} \ t_{vc} \ t_{pk} \ t_{vk} \ t_{upck} \ t_{ck}],$$

Table 7. A basis for L expressed as the corresponding estimable functions with zero estimates.

$-P + CK + C + VK$
$PCK + VPCK - VPC + VPK$
$VPC + K - VK + VPK$
$VP + 2CK + 2VCK + PCK$
$3P + 3VP + C + VPK$
$-3VC - 2K + VPK$
$PC - 3K + 2PK$
$C - 2VC - 2PC - PK$
$CK - PCK + 3VC + 3PK$
$-3CK - 3VCK + C - K$
$VCK + PCK - 2C + 2PK$
$V + 3VP + PCK - 3VPCK$
$-362\mu + 6125V + 6487P$

where the subscripts correspond to the relevant treatment combinations in the order corresponding to $\underline{y}$. The theoretical results necessary for this approach are given in Appendix 3. In particular the subspace L is isomorphic to the subspace of vectors

$$V = \{\underline{v} : N\underline{v} = \underline{0}, \text{ and } \underline{v}'\underline{t} = 0\},$$

where $\underline{v}'TZ\underline{\beta} = \underline{\theta}'\underline{\beta}$ (estimable) and

$$(3.7) \quad N = \begin{bmatrix} \underline{n}'_1 \\ \underline{n}'_2 \end{bmatrix} = \begin{bmatrix} + & + & + & + & + & + & + & - & - & - & - & - & - & - \\ + & 0 & 0 & 0 & 0 & 0 & - & 0 & 0 & 0 & + & 0 & - & 0 & 0 & 0 \end{bmatrix}$$

(Corollary 3.1.3 of Appendix 3). Now a basis for V can be readily obtained from the basis for L expressed as estimable functions in Table 7 by observing that, if $\underline{\theta}'\underline{\beta}^0 = (\underline{0}' \quad \underline{\theta}')$ $S^{g_1} \begin{bmatrix} W' \\ Z' \end{bmatrix} \underline{y} = 0$, then $\underline{v}'\underline{t} = 0$ where $\underline{v}'$ can be taken as $\frac{1}{2} [\underline{0}' \quad \underline{\theta}'] S^{g_1} \begin{bmatrix} W' \\ Z' \end{bmatrix} T'$ (cf. the proof of Corollary 3.1.3 of Appendix 3).

The resulting basis does not, however, have a simple form. A more attractive basis for $\mathbf{V}$ was obtained by means of a computer-based search for linearly independent vectors, $\underline{v} \in \mathbf{V}$, which have a minimum number of non-zero coefficients equal to +1 and -1. This search yielded twelve such linearly independent vectors, $\underline{v}$, and the corresponding functions, $\underline{v}'\underline{t}$, are displayed as the first twelve entries of Table 8.

Table 8. A basis for $\mathbf{V}$ expressed as the corresponding set of linear functions $\underline{v}'\underline{t}$.

$t_v - t_{vpc} - t_{vc} + t_{vpc}k$
$t_{vpc} - t_{vpk} + t_{(1)} - t_{vc}$
$t_{pk^*} - t_v - t_c + t_k - t_{pc} + t_{ck}$
$t_{pk^*} - t_{vpc} - t_c + t_{pc^*} - t_{(1)} + t_{vk}$
$t_{pk^*} - t_c - t_{vpk} + t_{vck} + t_{pk} - t_{ck}$
$t_{pk^*} - t_k - t_{vp} - t_{(1)} + t_{pk} + t_{vpc}k$
$t_v - t_{vpc} - t_c + t_{vpk} - t_{vp} + t_{(1)}$
$t_v - t_{vpc} + t_c - t_k - t_{(1)} + t_{ck}$
$t_v - t_c + t_k - t_{vck} - t_{vp} + t_{vk}$
$t_v - t_{vpk} + t_{pc^*} - t_{vck} - t_{pk} + t_{vpc}k$
$t_{pk^*} - t_v + t_{vpc} - t_c + t_{vp} - t_{vc} + t_{pk} - t_{vk}$
$t_{pk^*} - t_v - t_{vpc} + t_c + t_{(1)} - t_{pc} + t_{vk} - t_{ck}$
$8t_c - 9t_{pc^*} - 9t_{pc} + 8t_{vk}$

To complete the basis for $\mathbf{V}$ a vector $\underline{v} \in \mathbf{V}$ with four non-zero entries between -9 and +9 was obtained by a systematic computer-based search and the corresponding function, $\underline{v}'\underline{t}$, is shown as the last entry in Table 8.

It is interesting to investigate whether or not such simple bases for $\mathbf{L}$ and $\mathbf{V}$ as those represented in Tables 7 and 8 respectively do in fact exist in general. A simulation study to estimate selected probabilities associated with such bases was therefore undertaken. The model (2.1) with $E(\underline{y}) = W\underline{\gamma} + Z\underline{\beta}$, in which $\underline{\gamma}$ and $\underline{\beta}$ were taken as $\hat{\underline{\gamma}}$ and $\underline{\beta}^0$ of the present example, was assumed to be the true one and σ^2 was taken to be the error mean square of Table 4, i.e. $\sigma^2 = 590.796$. It was further assumed that the errors of (2.1) were normally distributed but, to comply

with the recorded results of the present experiment, the observations were rounded-off to the nearest integer. Then 1000 sets of 32 observations were generated according to this scheme. The probabilities of obtaining any of the individual null relations $\underline{\theta}'\underline{\beta}^0 = 0$ of Table 7 or $\underline{v}'\underline{t} = 0$ of Table 8 did not differ significantly from zero. However the probability of observing at least one estimable function, $\underline{\theta}'\underline{\beta}$, with zero estimate, where $\underline{\theta}$ has exactly four non-zero entries equal to +1 or -1, was estimated as 0.860 (± 0.011), while the probability of observing at least one relation $\underline{v}'\underline{t}$ where $\underline{v} \in \mathbf{V}$ has exactly four non-zero entries equal to +1 or -1 was estimated as 0.965 (± 0.006). Thus overall it is clear that, provided yields are recorded to the nearest integer, the probabilities of obtaining simple null relationships $\underline{\theta}'\underline{\beta}^0 = 0$ where $\underline{\theta} \in \mathbf{L}$, and $\underline{v}'\underline{t} = 0$ where $\underline{v} \in \mathbf{V}$, are high.

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APPENDIX 1

Consider the linear model not of full rank written as

$$(A1.1) \quad \underline{y} = X\underline{\beta} + \underline{\varepsilon},$$

where $\underline{y}$ is an $n \times 1$ vector of observations, X of order $n \times p$ and rank $r < p$ is a matrix of fixed values, $\underline{\beta}$ is a $p \times 1$ vector of unknown parameters, and $\underline{\varepsilon}$ is an $n \times 1$ vector of random components such that $E(\underline{\varepsilon}) = 0$ and $E(\underline{\varepsilon}\underline{\varepsilon}') = \sigma^2 I$. The normal equations are given by

$$(A1.2) \quad X'X\underline{\beta}^0 = X'y,$$

where $\underline{\beta}^0$ is any solution.

Theorem 1.1 A necessary and sufficient condition for $\underline{\theta}'\underline{\beta}$ to be estimable is that $\underline{\theta}' \in \mathcal{R}(X)$ (Pringle and Rayner, 1971, pp. 80-81) or, equivalently, that $\underline{\theta}'C'_0 = \underline{0}'$ where C_0 is an orthogonal complement of X .

Definition 1.1 A set of parameters is said to be aliased if and only if there exists a linear relationship across the corresponding columns of X . The aliasing (Payne et al., 1987, p.319) is completely specified by

$$(A1.3) \quad XC'_0 = 0$$

Theorem 1.2 An individual parameter β_i is estimable if and only if it is not involved in aliasing.

Proof: From Theorem 1.1, an individual parameter β_i is estimable if and only if the i^{th} row of C'_0 is $\underline{0}'$, and thus, from (A1.3), if and only if β_i is not involved in aliasing.

Theorem 1.3 (Haines and Rayner, 1987). If $\underline{\beta}_\star^0$ is any solution to the normal equations (A1.2), then the solution $\underline{\beta}^0$ subject to the constraint $C\underline{\beta}^0 = \underline{c}$, where C is a matrix complementary to X , is given by

$$\underline{\beta}^0 = \underline{\beta}_\star^0 - C'_0(CC'_0)^{g_1}(C\underline{\beta}_\star^0 - \underline{c}),$$

where g_1 represents any generalized inverse.

APPENDIX 2

Dr. D. Livingstone (1955) gave an alternative version of the mathematical solution of section 2.4:

Consider Z_c partitioned as $\begin{bmatrix} Z^\dagger \\ Z_0 \end{bmatrix}$, i.e. $Z^\dagger$ is the matrix Z with the two rows of Z_a omitted. Now delete two columns of Z_c in such a way that the 2×2 matrix of elements common to Z_0 and the deleted columns has a non-zero determinant. Then, if the treatment effects corresponding to the deleted columns of Z_c are excluded from model (2.1), the resultant reduced

model has full rank (cf. Banerjee and Federer (1966)). (The presence of repeated rows in Z does not affect the result.)

APPENDIX 3

Theorem 3.1 For the model (A1.1) partitioned as $\underline{y} = X_1\underline{\beta}_1 + X_2\underline{\beta}_2 + \underline{\varepsilon}$, necessary and sufficient conditions for $\underline{d}'\underline{y}$ to be the (unique) best linear unbiased estimator (BLUE) of an estimable function $\underline{\theta}'\underline{\beta}_2$ are given by

$$(A3.1) \quad \begin{cases} (i) & \underline{d}'X_1 = \underline{0}' \quad \text{and} \quad \underline{d}'X_2 = \underline{\theta}' \\ (ii) & \underline{d} \in \mathcal{C}(X), \text{ where } \mathcal{C} \text{ denotes column space.} \end{cases}$$

Proof: For unbiasedness $E(\underline{d}'\underline{y}) = \underline{d}'X_1\underline{\beta}_1 + \underline{d}'X_2\underline{\beta}_2 \equiv \underline{\theta}'\underline{\beta}_2$, i.e. (A3.1)(i) is true. Conversely, if (A3.1)(i) is true, $E(\underline{d}'\underline{y}) = \underline{\theta}'\underline{\beta}_2$. A necessary and sufficient condition for an unbiased $\underline{d}'\underline{y}$ to have minimum variance, viz. $\underline{d} \in \mathcal{C}(X)$, and the uniqueness of $\underline{d}$ follow from linear model theory, for example from Theorems 6.2 and 6.3, respectively, of Pringle and Rayner (1971, pp. 81-82).

Note: $\underline{d}'X_2 = \underline{\theta}'$ can be shown to be equivalent to $\underline{d}'X_2 = \underline{\theta}'A^{g_1}A = \underline{\theta}'$, where $A = (I - X_1X_1^{g_1})X_2$, which in turn is equivalent to $\underline{\theta}'(X_2^*)^{g_1}X_2^* = \underline{\theta}'$, where $X_2^* = \{I - X_1(X_1'X_1)^{g_1}X_1'\}X_2$ and $(X_2^*)'X_2^* = X_2'X_2 - X_2'X_1(X_1'X_1)^{g_1}X_1'X_2$. There is therefore no contradiction between (A3.1) and Theorem 6.12 of Pringle and Rayner (1971).

Corollary 3.1.1 For model (2.1) $\underline{d}'\underline{y}$ is the BLUE of an estimable function $\underline{\theta}'\underline{\beta}$ if and only if $\underline{d}$ satisfies the conditions

$$(A3.2) \quad \begin{cases} (i) & \underline{d}'\underline{w}_{(3)} = 0 \quad \text{and} \quad \underline{d}'Z = \underline{\theta}' \\ (ii) & \underline{d} \in \mathcal{C}[\underline{w}_{(3)} \ Z] \end{cases}$$

where $\underline{w}_{(i)} (i = 1, 2, 3)$ denotes the i^{th} column of W .

Proof: Here X becomes $[W \ Z]$, $\underline{\beta}$ becomes $\begin{bmatrix} \underline{\gamma} \\ \underline{\beta} \end{bmatrix}$, and, as was seen in section 2.2, a necessary and sufficient condition for estimability of $\underline{\theta}'\underline{\beta}$ is $\underline{\theta}' \in \mathcal{R}(Z)$. From (A3.1)

$$(A3.3) \quad \begin{cases} (i) & \underline{d}'W = \underline{0}' \text{ and } \underline{d}'Z = \underline{\theta}' \\ (ii) & \underline{d} \in C[W \ Z]. \end{cases}$$

In addition we see from the matrix S given in Table 2 that

$$(A3.4) \quad \begin{cases} \underline{w}'_{(i)}\underline{w}_{(j)} = 0 & \text{for all } i \neq j \\ Z\underline{w}_{(i)} = \underline{0} & i = 1, 2. \end{cases}$$

Now assume (A3.3) is true. Then from (A3.3)(ii) $\underline{d}$ is expressible as $\sum_{i=1}^3 \lambda_i \underline{w}_{(i)} + Z\underline{\lambda}_4$, where λ_i ($i = 1, 2, 3$) and $\underline{\lambda}_4$ are appropriate coefficients, and from (A3.4) $\underline{d}'\underline{w}_{(i)} = \lambda_i \underline{w}'_{(i)}\underline{w}_{(i)}$ for $i = 1, 2$. Thus from (A3.3)(i) $\lambda_1 = \lambda_2 = 0$, so that $\underline{d} \in C[\underline{w}_{(3)} \ Z]$; also (A3.3)(i) obviously implies (A3.2)(i).

Conversely, assume that (A3.2) is true. Then $\underline{d}$ can be expressed as $\lambda_3 \underline{w}_{(3)} + Z\underline{\lambda}_4$ where λ_3 and $\underline{\lambda}_4$ are appropriate coefficients and thus, from (A3.4), it follows that $\underline{d}'\underline{w}_{(i)} = 0$ for $i = 1, 2$. Furthermore, by assumption $\underline{d}'\underline{w}_{(3)} = 0$, so that overall $\underline{d}'W = \underline{0}'$. Also if $\underline{d} \in C[\underline{w}_{(3)} \ Z]$ then clearly $\underline{d} \in C[W \ Z]$.

Corollary 3.1.2 For model (2.1), $\underline{v}'\underline{t}$, where $\underline{t}$ is the vector of treatment totals defined in section 3.3, is the BLUE of an estimable function $\underline{\theta}'\underline{\beta}$ if and only if $\underline{v}$ satisfies the conditions

$$(A3.5) \quad \begin{cases} (i) & \underline{v}'T\underline{w}_{(3)} = 0 \text{ and } \underline{v}'TZ = \underline{\theta}' \\ (ii) & \underline{v} \in C[T\underline{w}_{(3)} \ TZ]. \end{cases}$$

Proof: Since $\underline{v}'\underline{t} = \underline{v}'T\underline{y}$, the necessary and sufficient conditions are (A3.2) with $\underline{d}' = \underline{v}'T$. This gives (A3.5)(i). Also since $T = [I_{16} \ I_{16}]$ of full row-rank, it possesses a right inverse $T'(TT')^{-1} = \frac{1}{2}T'$ (Pringle and Rayner, 1971, p.28), so that $\underline{v} = \frac{1}{2}T\underline{d}$ and therefore (A3.5)(ii) is equivalent to (A3.2)(ii).

Corollary 3.1.3 Conditions (A3.5) in Corollary 3.1.2 are equivalent to $\underline{v}'TZ = \underline{\theta}'$ and $N\underline{v} = \underline{0}$, where N is given by (3.7).

Proof: First, since $\underline{n}_1 = \frac{1}{2}T\underline{w}_{(3)}$, $\underline{v}'T\underline{w}_{(3)} = 0$ if and only if $\underline{v}'\underline{n}_1 = 0$.

Secondly, $\underline{n}_2$ is an orthogonal complement of $[T\underline{w}_{(3)} \ TZ]$, so that $\underline{v}'\underline{n}_2 = 0$ is equivalent to (A3.5)(ii). This follows since $\underline{n}_2'TZ$ represents the contrasts $pk^* - pk$ and $pc^* - pc$ corresponding to two sets of identical rows of TZ , while $\underline{n}_2$ is also orthogonal to $T\underline{w}_{(3)} = 2[\underline{a} \ -\underline{a}]'$.